5.1 Structure of a C Program

A C program consists of header files, constant definitions, function prototypes, the "main()" function, and other functions. Header files help organise reusable declarations.

5.2 Include Guards

Include guards use "#ifndef", "#define", and "#endif" to prevent a header file from being included multiple times in the same translation unit.

5.3 Conditional Compilation

The directive "#ifdef DEBUG" enables debugging statements only when the program is compiled with "-DDEBUG". This helps display intermediate values during testing.

5.4 Difference Between Include Forms

"#include <file.h>" searches standard system include locations. "#include "file.h"" searches the current source file's directory first, then other configured include locations.

5.5 Alternative Concept and Complexity Analysis

Using "#ifdef DEBUG" removes debugging statements during compilation when DEBUG is not defined. Using a run-time variable keeps the debugging code in the executable and checks the variable whenever the relevant statement is reached. Conditional compilation can therefore avoid unnecessary run-time checks and reduce executable size.

```c
#include <stdio.h>

#define SLAB1_UNITS 100
#define RATE2_PAISE 225
#define RATE3_PAISE 450
#define FIXED_CHARGE_PAISE 5000L

long compute_bill(unsigned int units)
{
    long paise = FIXED_CHARGE_PAISE;

#ifdef DEBUG
    printf("Debug: Calculating bill for %u units\n", units);
#endif

    if (units > SLAB1_UNITS) {
        if (units <= 200) {
            paise += (long)(units - 100) * RATE2_PAISE;
        } else {
            paise += 100L * RATE2_PAISE;
            paise += (long)(units - 200) * RATE3_PAISE;
        }
    }

    return paise;
}

int main(void)
```

```
{
    unsigned int units = 150;
    long bill = compute_bill(units);

    printf("Units consumed: %u\n", units);
    printf("Total bill: Rs %ld.%02ld\n",
           bill / 100, bill % 100);

    return 0;
}
```

